

Lesson 7

Automating Dashboards in Power BI

Understand the Power BI data lifecycle from source to dashboard sharing

March 2025

Lesson Objectives

By the end of this lesson, students will be able to:

1. Understand the Power BI data lifecycle from connection to dashboard sharing
2. Evaluate their work environment and choose the best automation strategy
3. Apply two practical automation pipelines based on real-world situations
4. Identify when to use Import, DirectQuery, or Dataflows
5. Set up Power BI refresh and data update automation based on business needs

Data Pipeline & Data Infrastructure in a Company.

how data flows from raw sources to business intelligence and AI-driven insights in a real-world enterprise setup

▷ Where does data come from?

1 Internal Data Sources

- CRM
- ERP
- Tnx database

2 External Data Sources

- APIs (Google Analytics, OpenWeather)
- Social Media Data

3 Machine-Generated Data

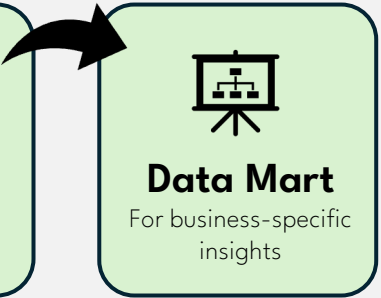
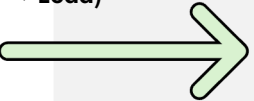
- IoT Sensors
- Server Logs & Web Scraping

4 Human-Generated Data

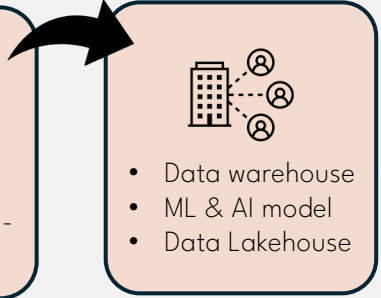
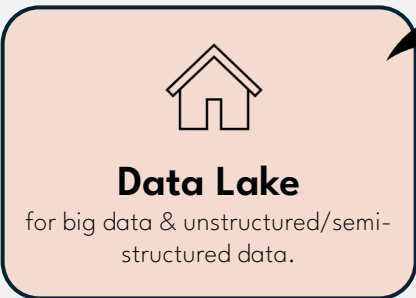
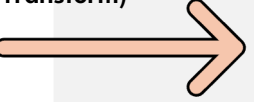
- Customer Feedback & Call Center Data
- Surveys & Forms

▷ How is data stored and processed?

ETL Pipeline
(Extract → Transform → Load)



ELT Pipeline
(Extract → Load → Transform)



▷ Who uses the processed data?

Business Intelligence (BI) & Reporting

- Dashboards & reports for historical trends & KPI monitoring.

Self-Service Analytics

- Querying Data Marts using SQL, Python.
- Generating custom reports, and ad-hoc analyses.

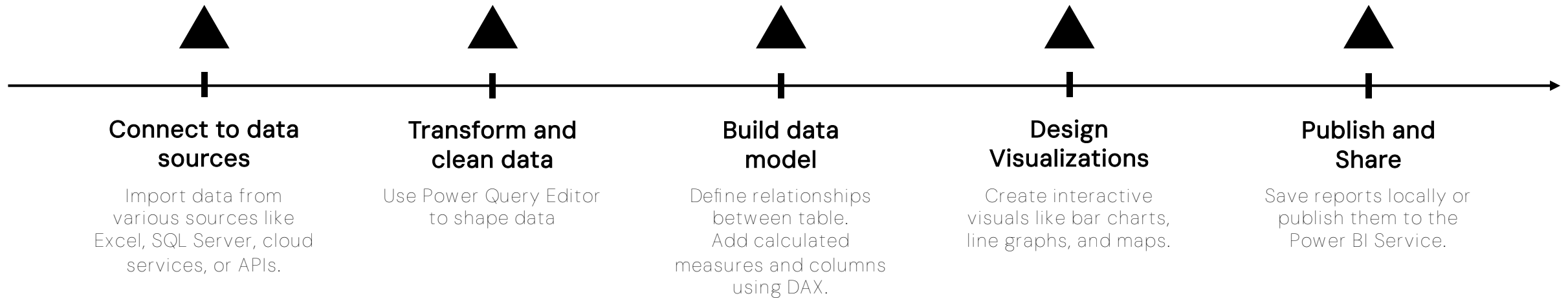
Machine Learning & AI-Driven Automation

- Running predictive models from Lake houses
- Automating decision-making (recommendation engines, fraud detection, dynamic pricing)

Power BI workflow.

From Data to Dashboard

Workflow

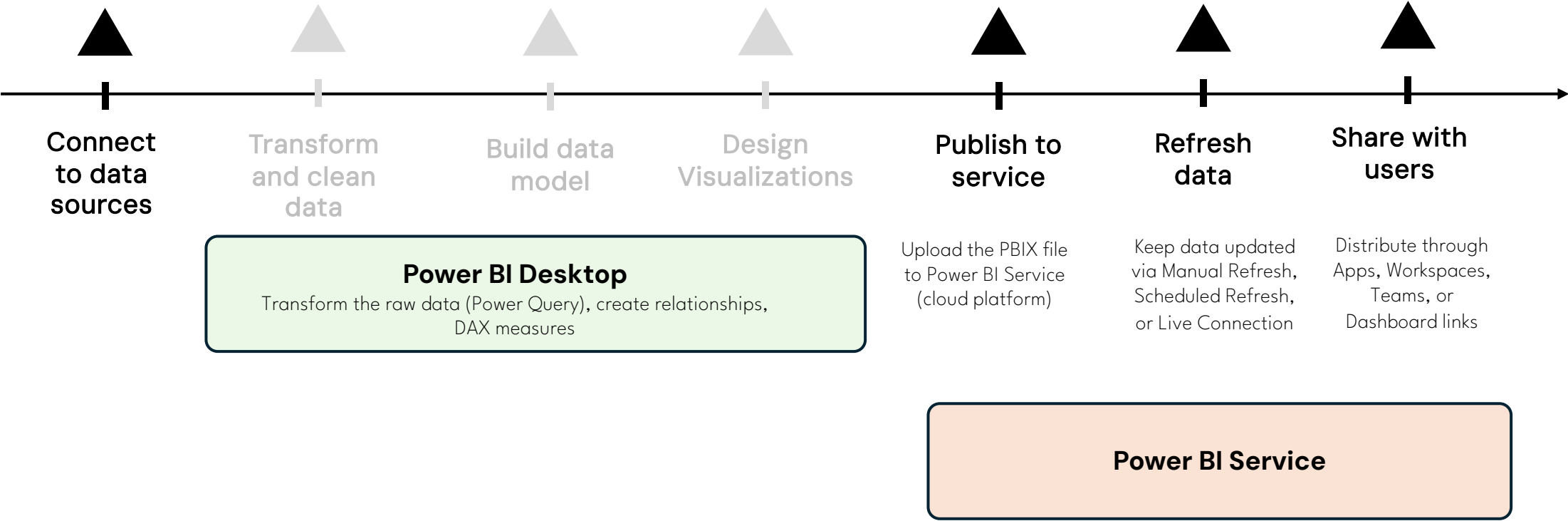


Connect → Transform → Model → Visualize → Publish

Power BI Desktop streamlines the process from raw data to actionable insights, enabling better decision-making.

Power BI Data Lifecycle.

from Source to Sharing



Import vs Direct Query vs Dataflow.

Choose the Right Connector!

Mode	Use When...	Pros	Cons
Import	Data doesn't need live updates	Fast performance, offline capable	Needs refresh schedule
Direct Query	Dashboard must reflect real-time	Always up-to-date	Slower visuals, depends on connection

Types of data sources & how they affect refresh.

Capabilities of your source determines whether you can use scheduled refresh

Source Type	Examples	Direct Query	Scheduled Refresh	Notes
Local Files	Excel, CSV, OneDrive (Personal)	✗ No	✗ No	Requires manual upload and re-publish
Org Cloud Storage	SharePoint, OneDrive Business	✓ Partial	✓ Yes	Most common solution for file-based team dashboards
Cloud Databases	Azure SQL, Snowflake	✓ Yes	✓ Yes	Needs Gateway or direct cloud connection
APIs & Web Services	REST API, Google Sheets	✗ No	✓ Yes (OAuth)	Requires custom connector or script to authenticate

Two Key Dimensions.

From Data to Dashboard



Working Context:

Who You Are / Where You Work

Case A: Analysts in a Company with Modern Data Infrastructure

→ There's a data warehouse, datamarts, and roles like DE/BI/DA are clearly defined.

Case B: Analysts in Teams Without a Data Platform

→ Must rely on external tools, CSV exports, OneDrive, Excel, etc.
→ They manage their own files and automation setups (e.g., OneDrive, Python)



Dashboard Needs & Data Volume:

Low or high

Low-volume / Simple Dashboards

→ Can work directly in Power BI using import mode

High-volume / Heavy Dashboards

→ Must be cleaned and aggregated before importing to avoid slow performance

Real-World Contexts for Automation

▷ Working Environment

Has Modern Data Infrastructure

- Dedicated Data Engineering team
- Centralized data via Data Warehouses (e.g., BigQuery, Snowflake)
- Automated ETL pipelines and Datamarts

→ **Power BI connects directly, updates in near real-time via Gateways or APIs**

No Infrastructure / Manual Setup

- Data stored in Excel, CSV, Google Sheets
- Users manually export reports from ERP, CRM, etc.
- Files saved on OneDrive, SharePoint, or local drives

→ **Everything managed through Power Query and manual or scheduled refresh**

▷ Data Form & Volume

Normal-sized Datasets

- Ranges from a few thousand to ~100,000 rows
- Daily or weekly updates
- Suitable for Excel/Sheets, small databases

→ **Easy to clean and transform using Power Query**

Massive-sized Datasets

- Over 1 million rows per day
- Requires near real-time updates
- Comes from POS systems, e-commerce platforms, or ERPs

→ **Requires optimized data modeling, incremental refresh, or DirectQuery**

The 2x2 Automation Strategy Matrix

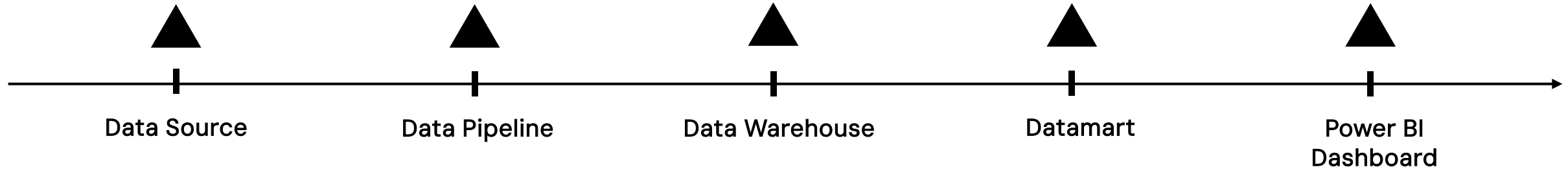
Helps you decide the correct pipeline depending on your context:

Environment	Data Size	Automation Strategy
Infra	Small/Medium	Power BI connects to Datamart or Warehouse → Scheduled Refresh
Infra	Large	DE builds optimized views → BI uses Direct Query or Aggregated Import
No Infra	Small/Medium	Manual data download → Save to OneDrive → Connect to Power BI → Auto Refresh
No Infra	Large	Python script for ETL → Save grouped file → Store in Cloud storage → Refresh

Automation Pipeline 1 - Company with Data Infrastructure

where Data Engineers manage pipelines and dashboards are powered by structured backend systems

Standardized Analytics Pipeline



The **Data Engineering (DE) team** uses the data pipeline to:

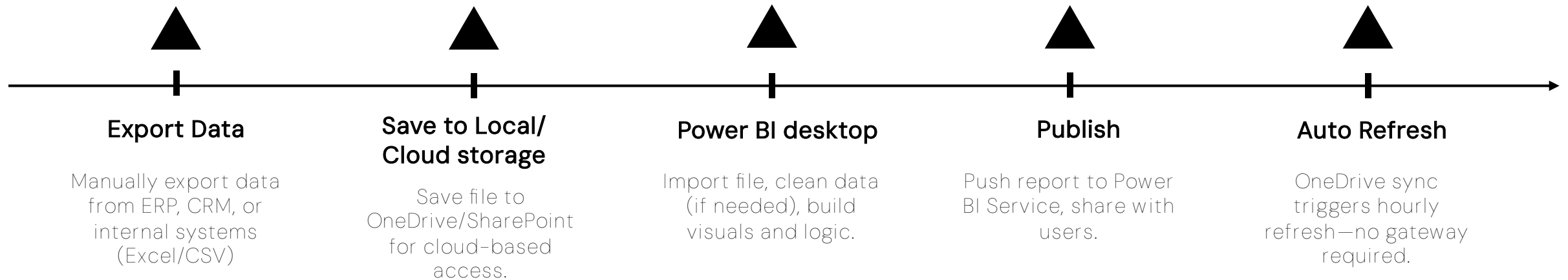
- Ingest data from various sources
- Clean, standardize, and validate the data
- Ensure schema consistency
- Load data into the Data Warehouse in an optimized, ready-to-use format

- **DA/ BI Developers** → Build reports directly from Datamarts, no raw data handling
- **Power BI System** → Uses Scheduled Refresh to keep dashboards up-to-date automatically

Automation Pipeline 2 – No Data Infrastructure

when analysts must self-manage data flow without support from a data platform

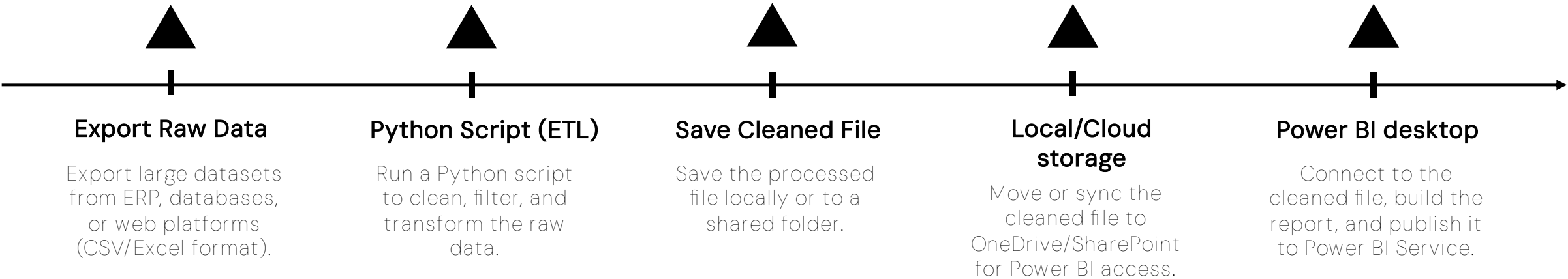
Flow for Small/Medium data size



Automation Pipeline 2 – No Data Infrastructure

when analysts must self-manage data flow without support from a data platform

Flow for Large data size

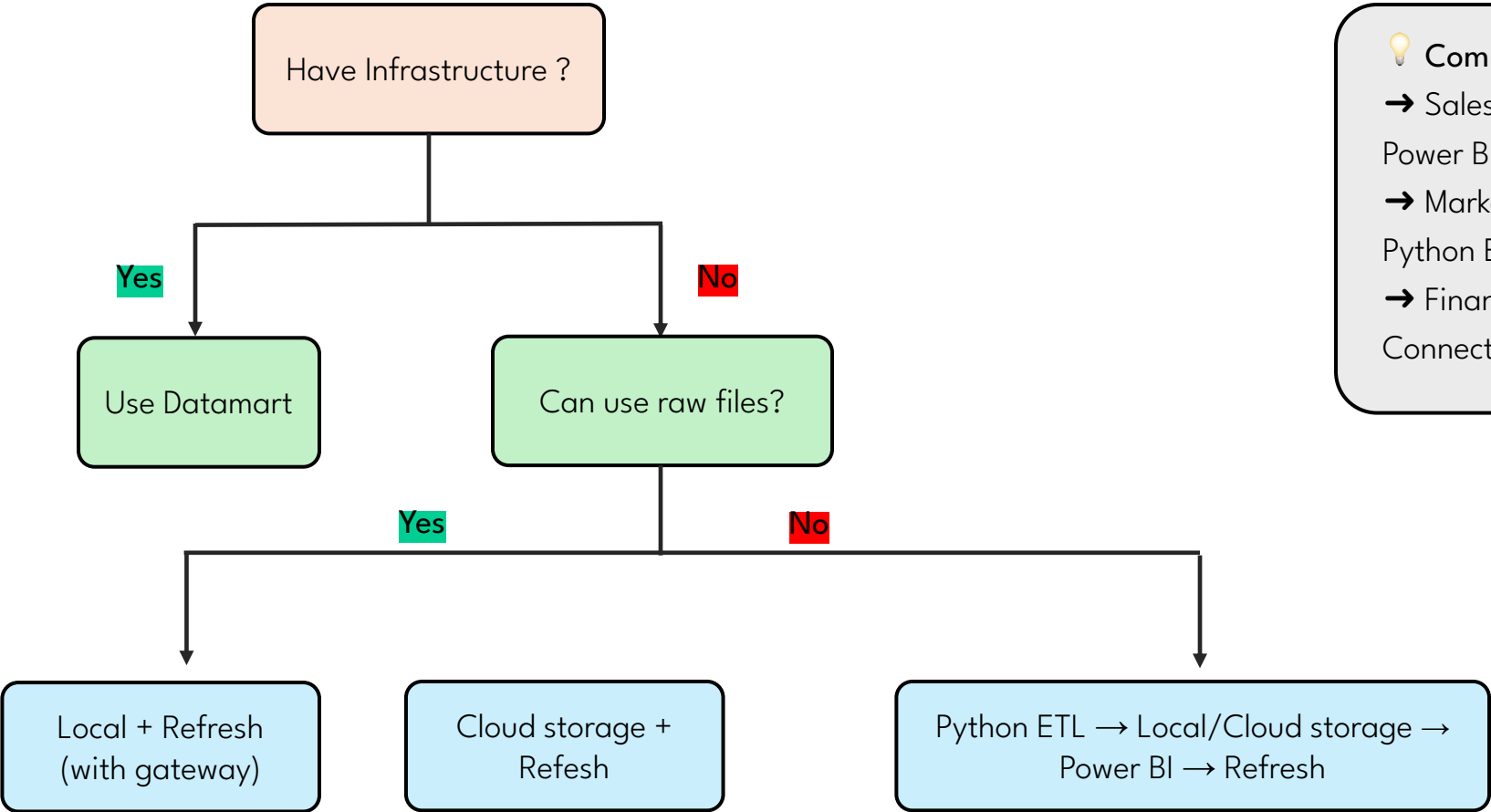


Automation tips

- Use **Task Scheduler** or **cron jobs** to automate the Python script
- Enable **Power BI Incremental Refresh** for large datasets if needed

Choosing the Right Automation Strategy

Use this quick logic tree to find the setup that fits your data environment.



- 💡 **Common Use Cases:**
- Sales Reports with low data volume → OneDrive + Power BI
 - Marketing data from Google Ads (large CSV) → Python ETL → Power BI
 - Finance team using company warehouse → Connect to Datamart